WATERSHED MODELING

USPCASW, MUET

FALL 2026

# RUNOFF MODELS



# RUNOFF MODELS

**Peak runoff models**

- Provide only the estimates of peak discharge from the watershed

**Continuous runoff models**

- This class of runoff models provides storm hydrographs for a given rainfall hyetograph
- Provide an estimate of runoff vs. time series

# PEAK RUNOFF MODELS

**Rational Method**

**NRCS Method**

# RATIONAL METHOD

To calculate peak runoff from small watersheds

Provides peak runoff rate from a catchment given:

- the runoff coefficient C,
- the time of concentration $T_c$,
- the area of the catchment, and
- the information to calculate the input or design storm or rainfall event

# RATIONAL METHOD: ASSUMPTIONS

- Catchment is small (less than 200 acres)
- Catchment is concentrated
- Rainfall intensity is uniform over the area of study
- The runoff coefficient is catch-all coefficient that incorporates all losses of the catchment

# RATIONAL METHOD: FORMULA

$$Qp = CI_iA$$

Qp = Peak discharge (cfs)

C = runoff coefficient (function of soil type, landuse/cover, and watershed slope)

$I_i$ = average rainfall intensity (in/hr) for a storm duration equal to the time of concentration, $T_c$

A = area (acre)

**For runoff coefficient:** refer to Bedient Table 6-5, page 381 (more detailed table in McCuen book --will be shared)

**TABLE 7–9** Runoff Coefficients for the Rational Formula versus Hydrologic Soil Group (A, B, C, D) and Slope Range

| | A | | | B | | | C | | | D | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Land Use | 0–2% | 2–6% | 6%+ | 0–2% | 2–6% | 6%+ | 0–2% | 2–6% | 6%+ | 0–2% | 2–6% | 6%+ |
| Cultivated land | 0.08[a] | 0.13 | 0.16 | 0.11 | 0.15 | 0.21 | 0.14 | 0.19 | 0.26 | 0.18 | 0.23 | 0.31 |
| | 0.14[b] | 0.18 | 0.22 | 0.16 | 0.21 | 0.28 | 0.20 | 0.25 | 0.34 | 0.24 | 0.29 | 0.41 |
| Pasture | 0.12 | 0.20 | 0.30 | 0.18 | 0.28 | 0.37 | 0.24 | 0.34 | 0.44 | 0.30 | 0.40 | 0.50 |
| | 0.15 | 0.25 | 0.37 | 0.23 | 0.34 | 0.45 | 0.30 | 0.42 | 0.52 | 0.37 | 0.50 | 0.62 |
| Meadow | 0.10 | 0.16 | 0.25 | 0.14 | 0.22 | 0.30 | 0.20 | 0.28 | 0.36 | 0.24 | 0.30 | 0.40 |
| | 0.14 | 0.22 | 0.30 | 0.20 | 0.28 | 0.37 | 0.26 | 0.35 | 0.44 | 0.30 | 0.40 | 0.50 |
| Forest | 0.05 | 0.08 | 0.11 | 0.08 | 0.11 | 0.14 | 0.10 | 0.13 | 0.16 | 0.12 | 0.16 | 0.20 |
| | 0.08 | 0.11 | 0.14 | 0.10 | 0.14 | 0.18 | 0.12 | 0.16 | 0.20 | 0.15 | 0.20 | 0.25 |
| Residential lot | 0.25 | 0.28 | 0.31 | 0.27 | 0.30 | 0.35 | 0.30 | 0.33 | 0.38 | 0.33 | 0.36 | 0.42 |
| size 1/8 acre | 0.33 | 0.37 | 0.40 | 0.35 | 0.39 | 0.44 | 0.38 | 0.42 | 0.49 | 0.41 | 0.45 | 0.54 |
| Residential lot | 0.22 | 0.26 | 0.29 | 0.24 | 0.29 | 0.33 | 0.27 | 0.31 | 0.36 | 0.30 | 0.34 | 0.40 |
| size 1/4 acre | 0.30 | 0.34 | 0.37 | 0.33 | 0.37 | 0.42 | 0.36 | 0.40 | 0.47 | 0.38 | 0.42 | 0.52 |
| Residential lot | 0.19 | 0.23 | 0.26 | 0.22 | 0.26 | 0.30 | 0.25 | 0.29 | 0.34 | 0.28 | 0.32 | 0.39 |
| size 1/3 acre | 0.28 | 0.32 | 0.35 | 0.30 | 0.35 | 0.39 | 0.33 | 0.38 | 0.45 | 0.36 | 0.40 | 0.50 |
| Residential lot | 0.16 | 0.20 | 0.24 | 0.19 | 0.23 | 0.28 | 0.22 | 0.27 | 0.32 | 0.26 | 0.30 | 0.37 |
| size 1/2 acre | 0.25 | 0.29 | 0.32 | 0.28 | 0.32 | 0.36 | 0.31 | 0.35 | 0.42 | 0.34 | 0.38 | 0.48 |
| Residential lot | 0.14 | 0.19 | 0.22 | 0.17 | 0.21 | 0.26 | 0.20 | 0.25 | 0.31 | 0.24 | 0.29 | 0.35 |
| size 1 acre | 0.22 | 0.26 | 0.29 | 0.24 | 0.28 | 0.34 | 0.28 | 0.32 | 0.40 | 0.31 | 0.35 | 0.46 |
| Industrial | 0.67 | 0.68 | 0.68 | 0.68 | 0.68 | 0.69 | 0.68 | 0.69 | 0.69 | 0.69 | 0.69 | 0.70 |
| | 0.85 | 0.85 | 0.86 | 0.85 | 0.86 | 0.86 | 0.86 | 0.86 | 0.87 | 0.86 | 0.86 | 0.88 |
| Commercial | 0.71 | 0.71 | 0.72 | 0.71 | 0.72 | 0.72 | 0.72 | 0.72 | 0.72 | 0.72 | 0.72 | 0.72 |
| | 0.88 | 0.88 | 0.89 | 0.89 | 0.89 | 0.89 | 0.89 | 0.89 | 0.90 | 0.89 | 0.89 | 0.90 |
| Streets | 0.70 | 0.71 | 0.72 | 0.71 | 0.72 | 0.74 | 0.72 | 0.73 | 0.76 | 0.73 | 0.75 | 0.78 |
| | 0.76 | 0.77 | 0.79 | 0.80 | 0.82 | 0.84 | 0.84 | 0.85 | 0.89 | 0.89 | 0.91 | 0.95 |
| Open space | 0.05 | 0.10 | 0.14 | 0.08 | 0.13 | 0.19 | 0.12 | 0.17 | 0.24 | 0.16 | 0.21 | 0.28 |
| | 0.11 | 0.16 | 0.20 | 0.14 | 0.19 | 0.26 | 0.18 | 0.23 | 0.32 | 0.22 | 0.27 | 0.39 |
| Parking | 0.85 | 0.86 | 0.87 | 0.85 | 0.86 | 0.87 | 0.85 | 0.86 | 0.87 | 0.85 | 0.86 | 0.87 |
| | 0.95 | 0.96 | 0.97 | 0.95 | 0.96 | 0.97 | 0.95 | 0.96 | 0.97 | 0.95 | 0.96 | 0.97 |

[a] Runoff coefficients for storm recurrence intervals less than 25 years.

[b] Runoff coefficients for storm recurrence intervals of 25 years or longer.

# Key Details of Table 7-9

**Parameters covered:**

- **Land Use / Cover Types:** such as agricultural, residential, commercial, industrial, or open/natural land covers like grasslands and forests.
- **Hydrologic Soil Groups (A, B, C, D):** Classifies soils based on infiltration and transmission rates (from high infiltration sand/gravel types in Group A to low infiltration clay types in Group D).
- **Slope Ranges:** Categorized by general gradient classes (e.g., flat, rolling, steep), which affect the velocity and volume of surface runoff.

**Application:** The table values represent empirical runoff coefficients (C) used in peak discharge equations.

# RUNOFF COEFFICIENT FOR NONHOMOGENEOUS AREA

Weighted runoff coefficient based on the area of each land use

$$Cw = \sum_{j=1}^{n} Cj\ Aj / \sum_{j=1}^{n} Aj$$

# NRCS RUNOFF CURVE NUMBER METHODS

- By the USA Soil Conservation Service (now called the Natural Resources Conservation Service), division of the USDA (US Department of Agriculture)
- To predict peak discharge due to a 24-hr storm event
- Empirically derived relationships that use precipitation, land cover, hydrologic soil group, and watershed characteristics to estimate runoff depth. When combined with a unit hydrograph method, it can also be used to estimate peak discharge and runoff hydrographs.
- More sophisticated approach than the Rational Method

# NRCS CURVE NUMBER

Curve number is a coefficient that reduces the total precipitation to runoff potential, after “losses”

- Evaporation
- Absorption
- Transpiration
- Surface Storage

*The higher the CN value, the higher the runoff potential will be*

It is essential to use the CN value that best mimics the **Ground Cover Type** and **Hydrologic Condition**

# NRCS RAINFALL-RUNOFF EQUATION

The following equation presents the relationship between accumulated rainfall and accumulated runoff

$$Q = \frac{(P - I_a)^2}{(P - I_a) + S}$$ Equation 1

Where:

Q = accumulated direct runoff (inches or mm)

P = accumulated rainfall (potential maximum runoff) (inches or mm)

$I_a$ = initial abstraction including surface storage, interception, evaporation, and infiltration prior to any runoff occurring (inches or mm)

S = potential maximum soil moisture retention after runoff begins (inches or mm)

Note: for $P \leq I_a$, $Q = 0$

# POTENTIAL MAXIMUM RETENTION (S)

Potential maximum retention (S) can be calculated using Equation 2.

$$S = z\left(\frac{100}{CN} - 1\right)$$

Equation 2

Where:

z=10 for English measurement units, or 254 for metric

CN = Runoff Curve Number (0 - 100)

Generally, $I_a$ may be estimated as $\mathbf{I_a = 0.2\ S}$ Equation 3

Substituting $I_a$ value in Equation 1

$$Q = \frac{(P - 0.2S)^2}{(P + 0.8S)}$$

Equation 4

# NRCS RUNOFF EQUATIONS

$$Q = \frac{(P - I_a)^2}{(P - I_a) + S} \qquad \text{[eq. 2-1]}$$

where

Q = runoff (in)
P = rainfall (in)
S = potential maximum retention after runoff begins (in) and
$I_a$ = initial abstraction (in)

$$I_a = 0.2S \qquad \text{[eq. 2-2]}$$

$$Q = \frac{(P - 0.2S)^2}{(P + 0.8S)} \qquad \text{[eq. 2-3]}$$

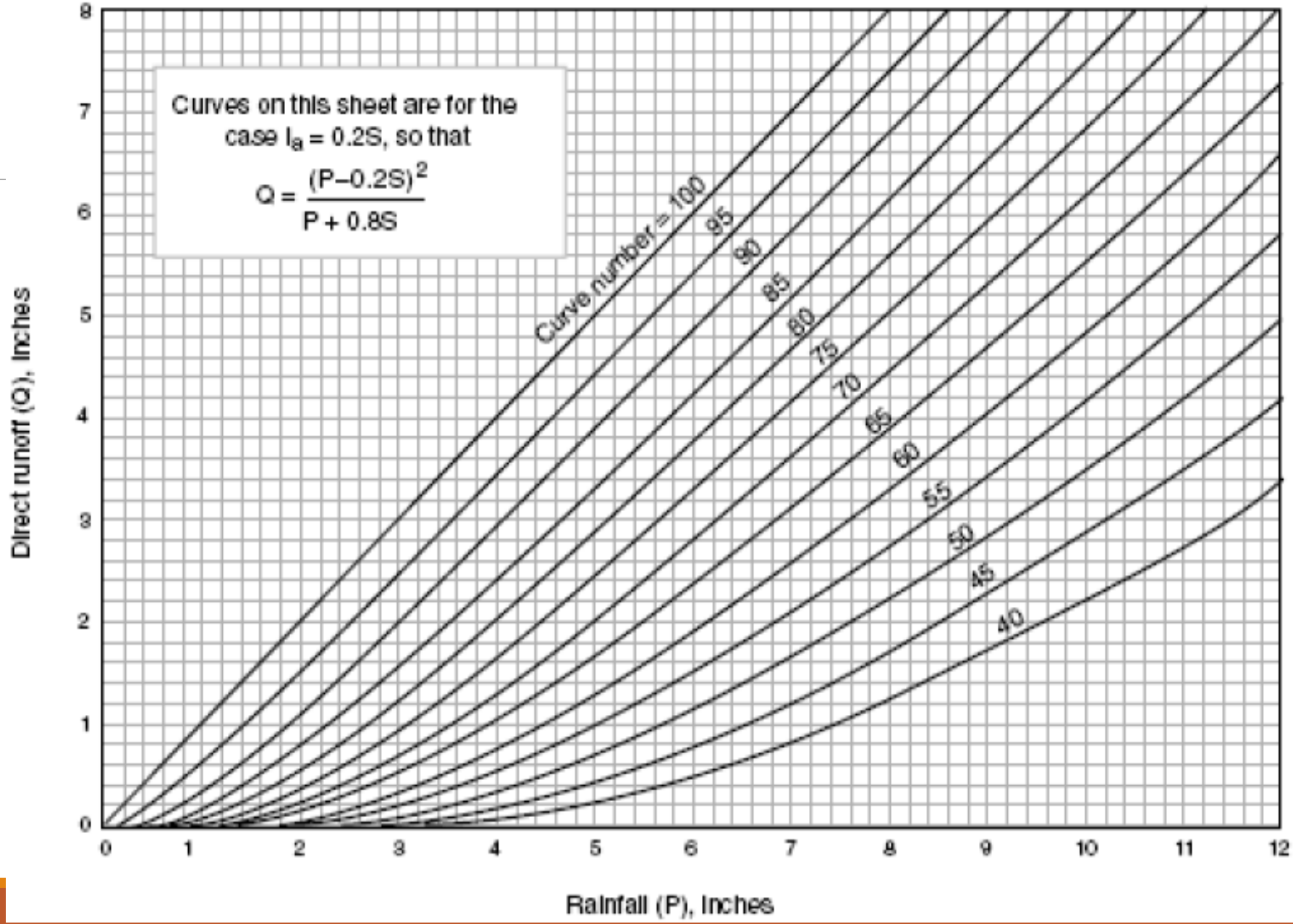
$$S = \frac{1000}{CN} - 10 \qquad \text{[eq. 2-4]}$$

Figure 2-1 (runoff equation):

$$Q = \frac{\left[P - .2\left(\frac{1000}{CN} - 10\right)\right]^2}{P + 0.8\left(\frac{1000}{CN} - 10\right)}$$

where
Q = runoff (in)
P = rainfall (in)
CN = runoff curve number

The solution to this equation results in runoff depth in "watershed" inches.


Curves on this sheet are for the case $I_a = 0.2S$, so that
$Q = \frac{(P-0.2S)^2}{P+0.8S}$
Curve number = 100
95
90
85
80
75
70
65
60
55
50
45
40
Direct runoff (Q), Inches
Rainfall (P), Inches
0
1
2
3
4
5
6
7
8
9
10
11
12

# Curve Number CN

The principal physical watershed characteristics affecting the relationship between rainfall and runoff are

- land use, land treatment,
- soil types, and
- land slope.

NRCS method uses a combination of soil conditions and land uses (ground cover) to assign a runoff factor to an area (CN).

CN indicates the runoff potential of an area

- The higher the CN, the higher the runoff potential

Soil properties also influence the relationship between runoff and rainfall since soils have differing rates of infiltration.

Based on infiltration rates, the NRCS has divided soils into four hydrologic soil groups.

# HYDROLOGIC SOIL GROUPS

Hydrologic Group is a grouping of soils that have similar runoff potential under similar storm and land cover conditions

**Group A Soils:** High infiltration (low runoff). Sand, loamy sand, or sandy loam. Infiltration rate > 0.3 inch/hr when wet.

**Group B Soils:** Moderate infiltration (moderate runoff). Silt loam or loam. Infiltration rate 0.15 to 0.3 inch/hr when wet.

**Group C Soils:** Low infiltration (moderate to high runoff). Sandy clay loam. Infiltration rate 0.05 to 0.15 inch/hr when wet.

**Group D Soils:** Very low infiltration (high runoff). Clay loam, silty clay loam, sandy clay, silty clay, or clay. Infiltration rate 0 to 0.05 inch/hr when wet.

Effects of Urbanization: Consider the effects of urbanization on the natural hydrologic soil group. For compacted soil, you should make appropriate changes to the selected soil group.

Antecedent soil moisture conditions: AMC I, II, and III

# ANTECEDENT SOIL MOISTURE CONDITIONS-AMC

AMC is the preceding relative moisture of the pervious surfaces prior to the rainfall event

- **Low**: when there has been little preceding rainfall
- **High:** when there has been considerable preceding rainfall prior to the modeled rainfall event

**ACM I (dry), ACM II (average), and ACM III (wet)**

**For modeling purposes, we consider watersheds to be AMC II, which is essentially an average moisture condition**

**For areas with high water table conditions, AMC III antecedent soil moisture conditions may be used.**

# CNs

| Land Use | Hydrologic Soil Group | | | |
|---|---|---|---|---|
| | A | B | C | D |
| Pre-Developed or Undisturbed Pervious | 55 | 71 | 80 | 84 |
| Developed or Disturbed Pervious | 71 | 80 | 84 | 88 |
| Impervious | 98 | 98 | 98 | 98 |

A CN of 100 is to be used for permanent water surfaces such as lakes and ponds

**Table 2:** NRCS runoff curve numbers (CN) for selected urban land use[1]

| *Cover description* | | *Curve numbers for hydrologic soil group* | | | |
|---|---|---|---|---|---|
| **Cover type and hydrologic condition** | **Average impervious area[2]** | **A** | **B** | **C** | **D** |
| *Fully developed urban areas (vegetation established)* | | | | | |
| Open space (lawns, parks, golf courses, cemeteries, etc)[3]: | | | | | |
| Poor condition (grass cover <50%) | | 68 | 79 | 86 | 89 |
| Fair condition (grass cover 50% to 75%) | | 49 | 69 | 79 | 84 |
| Good condition (grass cover >75%) | | 39 | 61 | 74 | 80 |
| Impervious areas: | | | | | |
| Paved parking lots, roofs, driveways, etc (excluding ROW) | | 98 | 98 | 98 | 98 |
| Streets and roads: | | | | | |
| Paved: curbs and storm sewers (excluding ROW) | | 98 | 98 | 98 | 98 |
| Paved: open ditches (including ROW) | | 83 | 89 | 92 | 93 |
| Gravel (including ROW) | | 76 | 85 | 89 | 91 |
| Dirt (including ROW) | | 72 | 82 | 87 | 89 |
| Western desert urban areas: | | | | | |
| Natural desert landscaping (pervious areas only)[4] | | 63 | 77 | 85 | 88 |
| Artificial desert landscaping (impervious weed barrier, desert shrub with 1-2 inch sand or gravel mulch and basin borders) | | 96 | 96 | 96 | 96 |
| Urban districts: | | | | | |
| Commercial and business | 85 | 89 | 92 | 94 | 95 |
| Industrial | 72 | 81 | 88 | 91 | 93 |
| Residential districts by average lot size: | | | | | |
| 1/8 acre or less (town houses) | 65 | 77 | 85 | 90 | 92 |
| 1/4 acre | 38 | 61 | 75 | 83 | 87 |
| 1/3 acre | 30 | 57 | 72 | 81 | 86 |
| 1/2 acre | 25 | 54 | 70 | 80 | 85 |
| 1 acre | 20 | 51 | 68 | 79 | 84 |
| 2 acres | 12 | 46 | 65 | 77 | 82 |
| *Developing urban areas* | | | | | |
| Newly graded areas (pervious areas only, no vegetation)[5] | 77 | 86 | 91 | 94 | |
| Idle lands (CN's are determined using cover types similar to those in Table 3) | | | | | |

1 Average runoff condition and $I_a$=0.2S.
2 The average percent impervious area shown was used to develop the composite CN's. Other assumptions are as follows: impervious areas are directly connected to the drainage system, impervious areas have a CN of 98, and pervious areas are considered equivalent to open space in good hydrologic condition. CN's for other combinations of conditions pay be computed using Figures 3 or 4.
3 CN's shown are equivalent to those of pasture. Composite CN's may be computed for other combinations of open space cover type.
4 Composite CN's for natural desert landscaping should be computed using Figures 3 or 4, based on the impervious area percentage (CN=98) and the pervious area CN. The pervious area CN's are assumed equivalent to desert shrub in poor hydrologic condition.
5 Composite CN's to use for the design of temporary measures during grading and construction should be computed using Figures 3 or 4 based on the degree of development (impervious area percentage) and the CN's for the newly graded pervious areas.

# COMPOSITE CURVE NUMBER

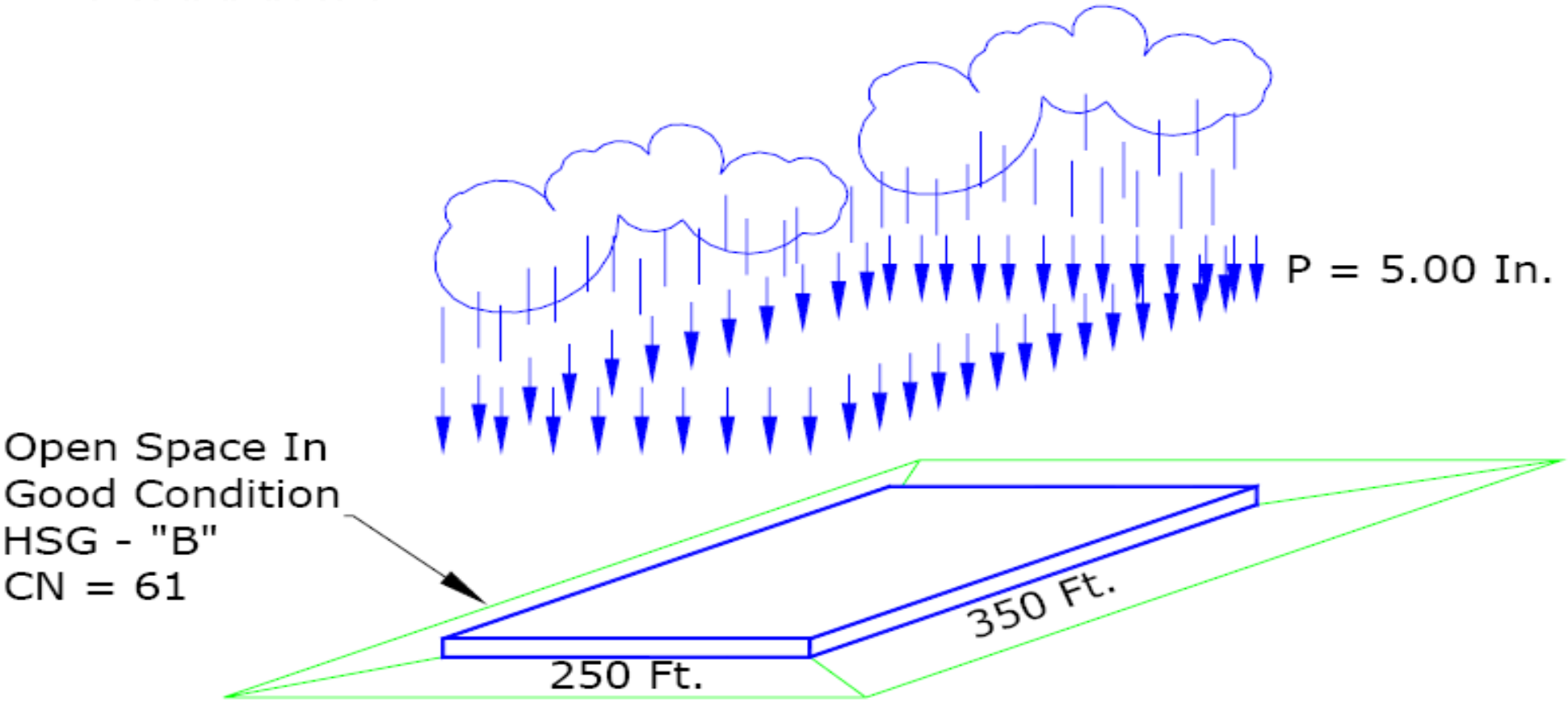
When a drainage area has more than one land use

When Composite CN is used

- The drainage area is considered a uniform land use represented by the composite curve number
- can be calculated by using the weighted method

| Composite Curve Number Calculation Example | | | |
|---|---|---|---|
| Land use | % of total land area | Curve number | Weighted curve number (% area x CN) |
| Residential 1/8-acre soil group B | 0.80 | 85 | 68 |
| Meadow good condition soil group C | 0.20 | 71 | 14 |
| **Total Weighted CN = 68 + 14 = 82** | | | |

# EXAMPLE




How much runoff volume would be expected from this circumstance ?

# Runoff Volume example (Continued):

1. Compute the Surface Storage:
   $S = (1000 / CN) - 10$
   $S = (1000 / 61) - 10 = 6.393$ Inches
2. Compute the Initial Abstraction:
   $I_a = 0.2 \times S$
   $I_a = 0.2 \times 6.393 = 1.279$ Inches
3. Compute the runoff in Watershed Inches:
   $Q = (P - I_a)^2 / (P - Ia + S)$
   $Q = (5.00 - 1.279)^2 / (5.00 - 1.279 + 6.393)$
   Q = **1.369 Inches** ***(Remember the original P=5.00 Inches)***
4. Compute the Runoff Volume:
   V = [1.369 In / (12 In / Ft)] x 250 Ft x 350 Ft =
   V = **9983 CF** or 9983 CF / (43560 SF / Ac) = **0.2293 Ac-Ft**

$V_{PRECIP}$ = [5.000 In / (12 In / Ft)] x 250 Ft x 350 Ft = 36,458 CF
Therefore the CN reduced the Precipitation Volume by 75%!

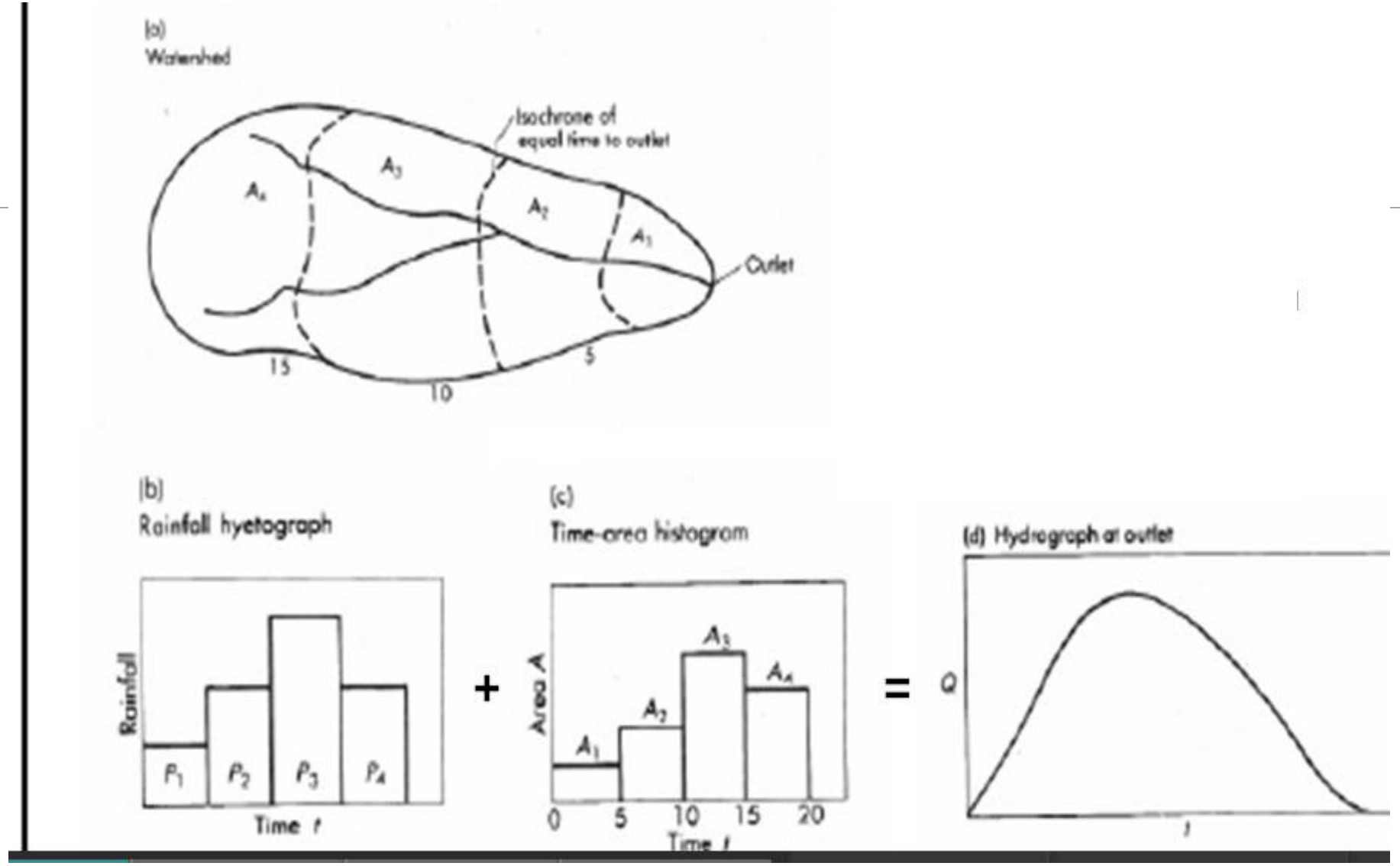
# CONTINUOUS RUNOFF MODELS

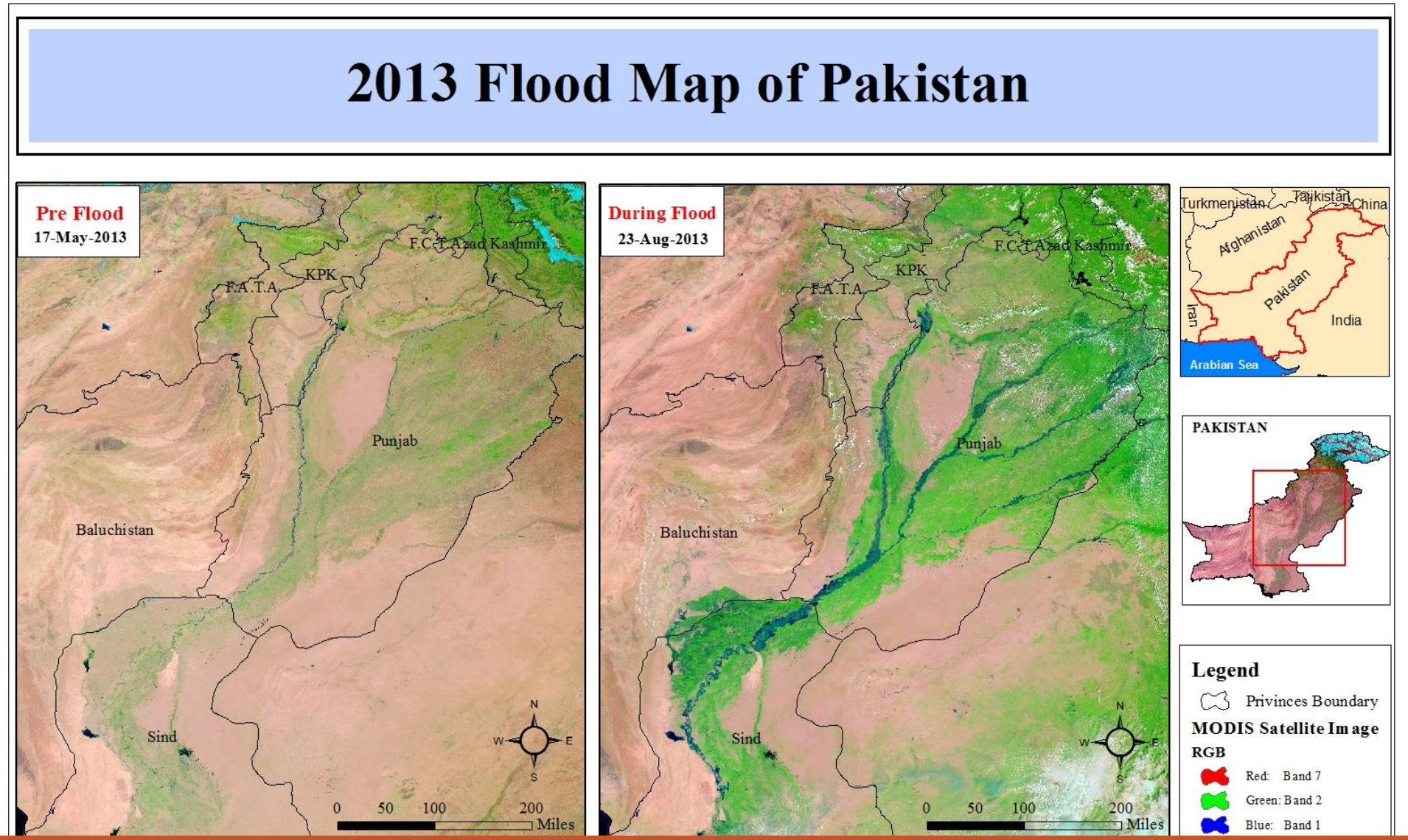
**Time area method**

**Unit Hydrograph Techniques**

# TIME AREA METHOD

- Develop to address non-uniform rainfall in large areas
- Convert rainfall excess into a hydrograph
- Concept of time-area histogram is used
- This method assumes that outflow hydrograph results from pure translation of direct runoff to the outlet at uniform velocity, ignoring any storage effects in the watershed
- Watershed is divided into subareas with distinct runoff translation times to the outlet
- Subareas are delineated with **isochrones of equal translation time** (numbered upstream from the outlet)


(a)
Watershed
Isochrone of
equal time to outlet
$A_4$
$A_3$
$A_2$
$A_1$
Outlet
15
10
5
(b)
Rainfall hyetograph
Rainfall
$P_1$
$P_2$
$P_3$
$P_4$
Time $t$
(c)
Time-area histogram
Area $A$
$A_1$
$A_2$
$A_3$
$A_4$
0
5
10
15
20
Time $t$
(d) Hydrograph at outlet
$Q$
$t$




# HYDROGRAPHS

Floods 2013 in Sindh

# FLOOD 2013 SITUATION IN SINDH

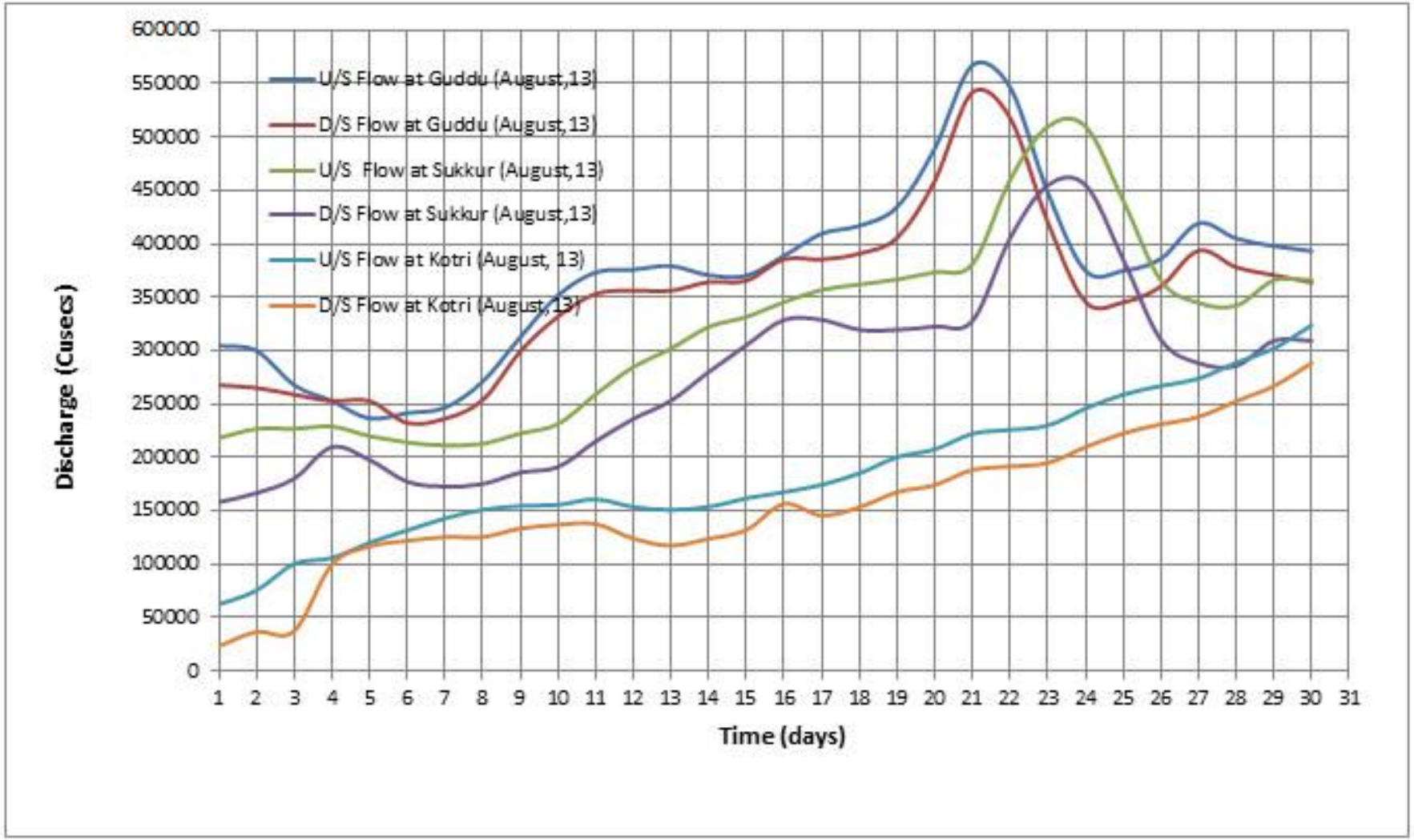



**Fig.1: shows Discharge at U/S and D/S of Guddu, Sukkur and Kotri barrages**

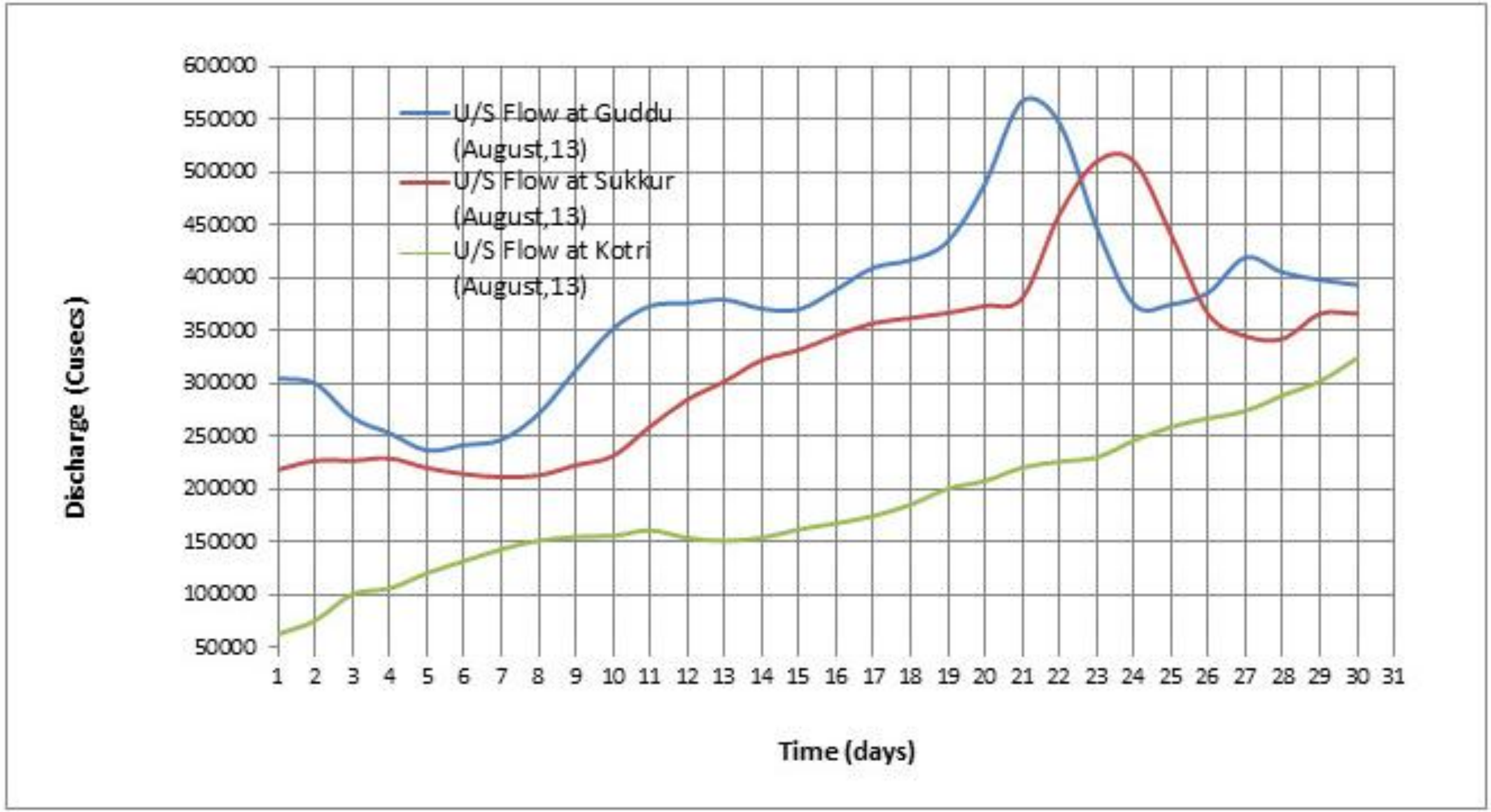



**Fig.2: shows Discharge at U/S Guddu , Sukkur and Kotri barrages**

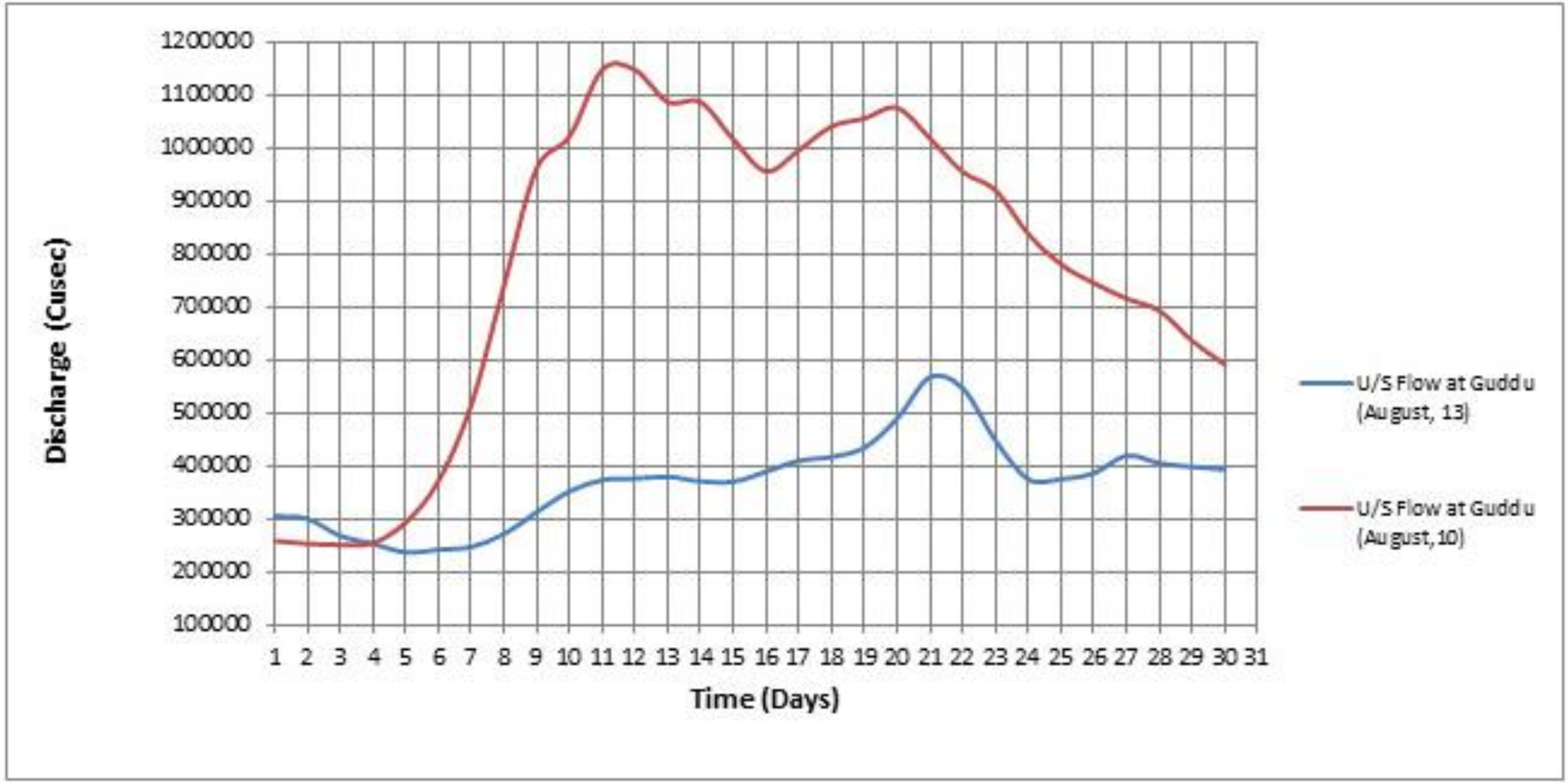



**Fig. 4: shows comparison of flood 2010 and 2013 at U/S Guddu**

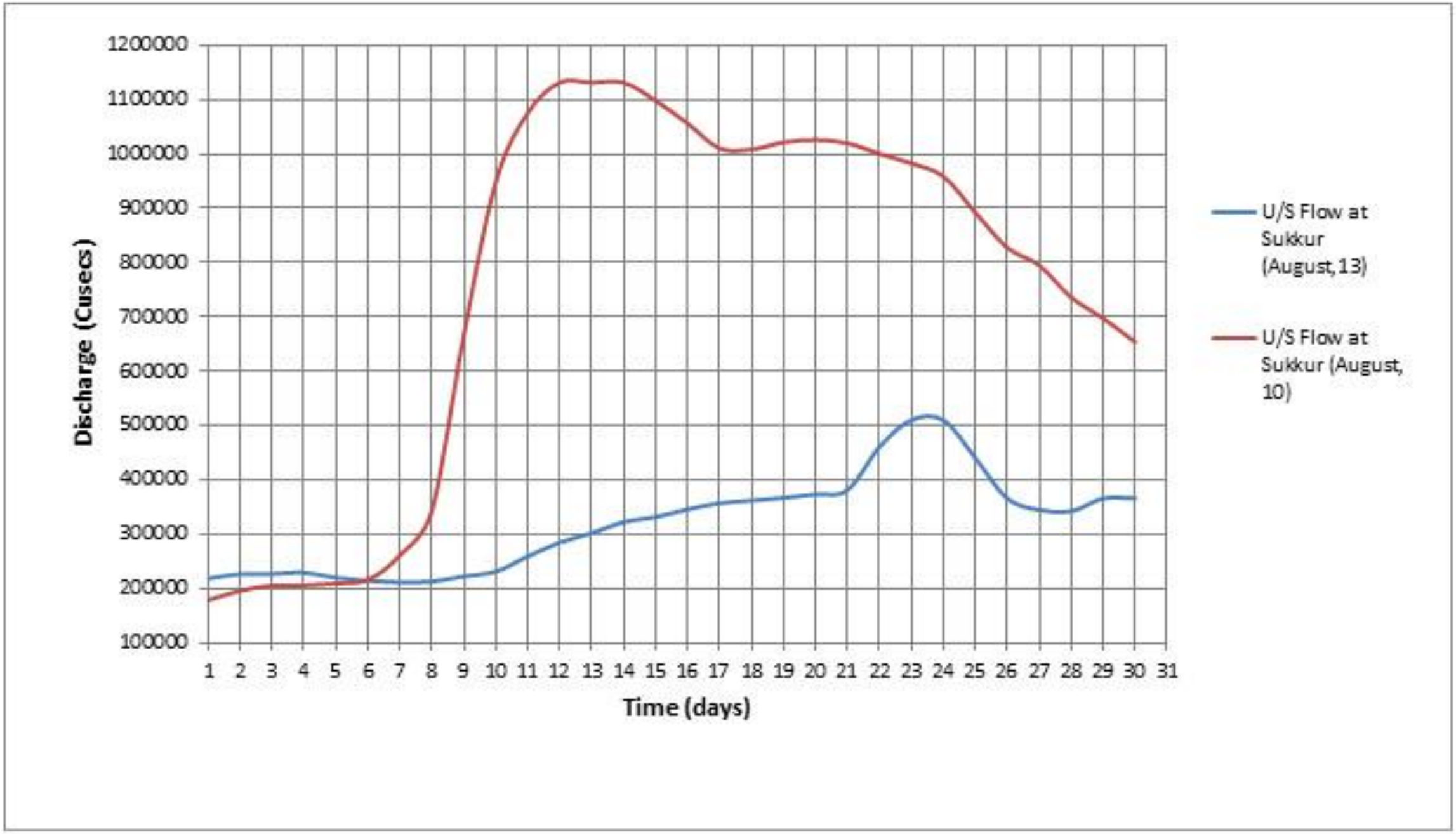



**Fig. 6: shows comparison of flood 2010 and 2013 at U/S, Sukkur**